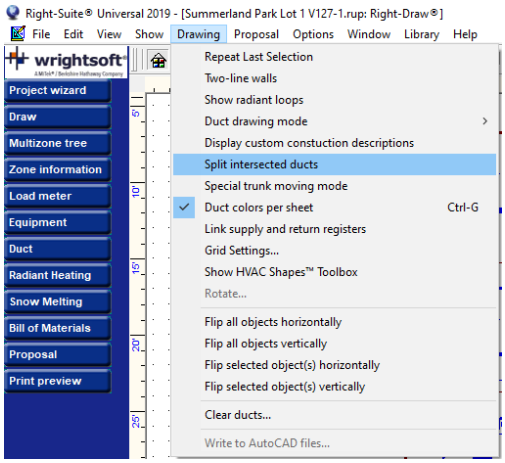
Rheia Duct Supplemental Guide

Section 1- Creating your Rheia template:

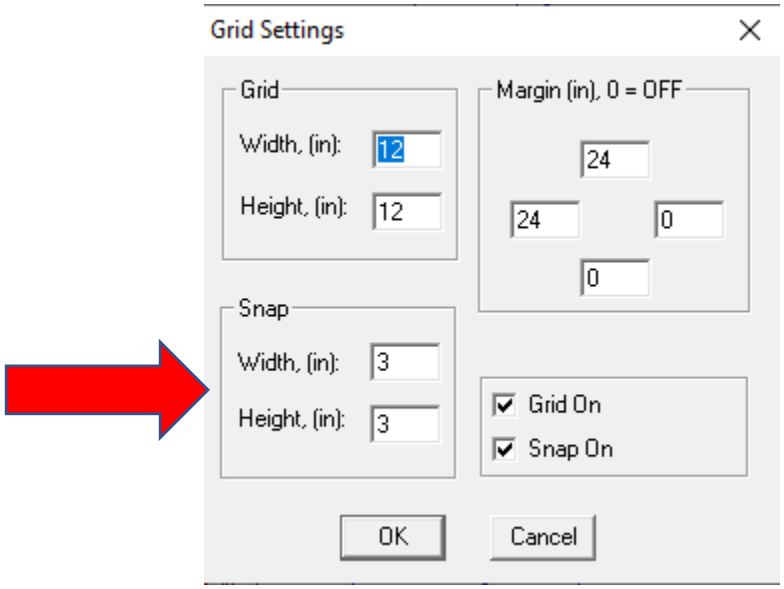
It is advised that users create a template for Rheia design projects. Some settings required for Rheia duct designs are ill-suited to standard duct designs and vice-versa.

Note: Standard settings used for a template (default or otherwise) will remain largely the same. Example: building materials, design temps, etc…

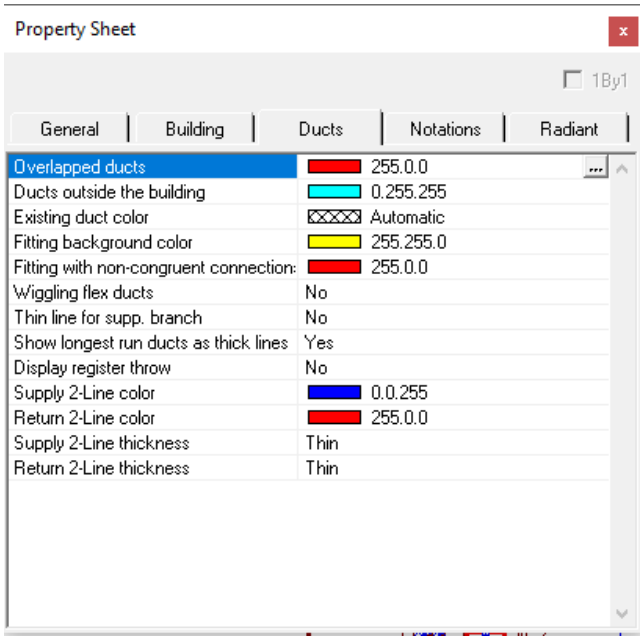
1. Open RSU as normal. This will be the setting from your default template (if this is the first time using RSU in any capacity follow the basic instructions for creating a standard default template first).
2. Go to the drawing screen. Go the 'Show" menu > select 'Right-Draw'
3. Go to the 'Drawing' menu. Select the 'Split intersected ducts' option **so that it is now unchecked**. This will prevent the wire helix duct from becoming interconnected.



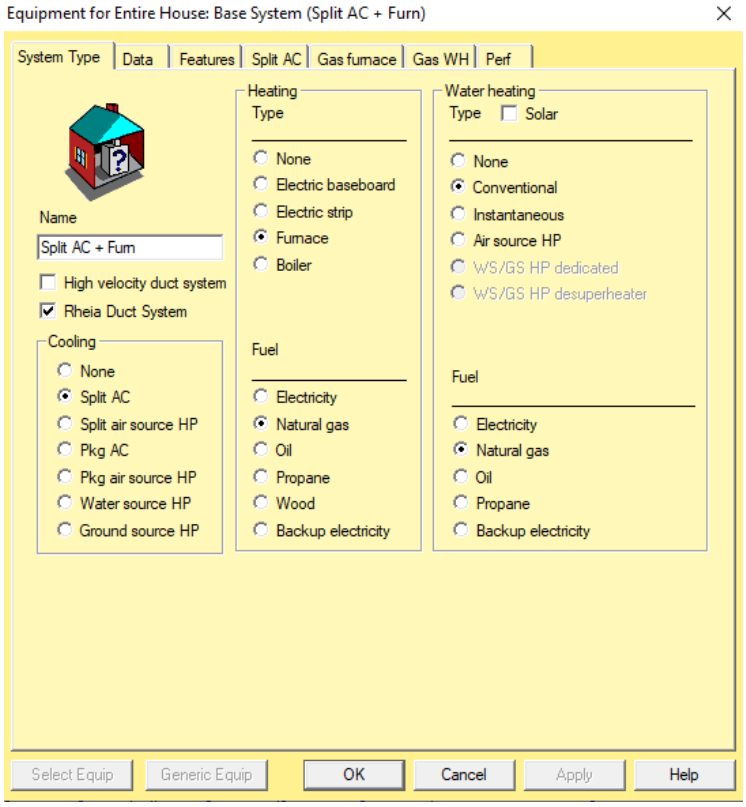
4. Return to the 'Drawing' menu. Select the 'Grid Settings' option. Change the Snap settings to 3" for both width and height. Leave the Snap 'On'. Do as you like with the grid. Use 'okay' to close the window.



5. While still on the drawing screen, right click in the blank drawing area. If you have not already done so, stretch out the property sheet window and adjust the location of the column divider to allow you to see the property options. Click on the 'Ducts' tab of the property sheet. Change the 'Wiggling flex ducts' option to 'No' and 'show longest ducts as thick lines' to 'yes' (this can be turned off for printing)
    a. If you do not see the 'Ducts' tab, leave the property sheet open and click anywhere in BLANK drawing area.



6. With the Drawing Screen property sheet still open, go to the 'Notations' tab and change the 'Display Riser Notation' to 'No' and close the property sheet.
7. Go to the 'Show' menu and select the equipment option. On the 'System Type' tab, select the 'Rheia System' box.

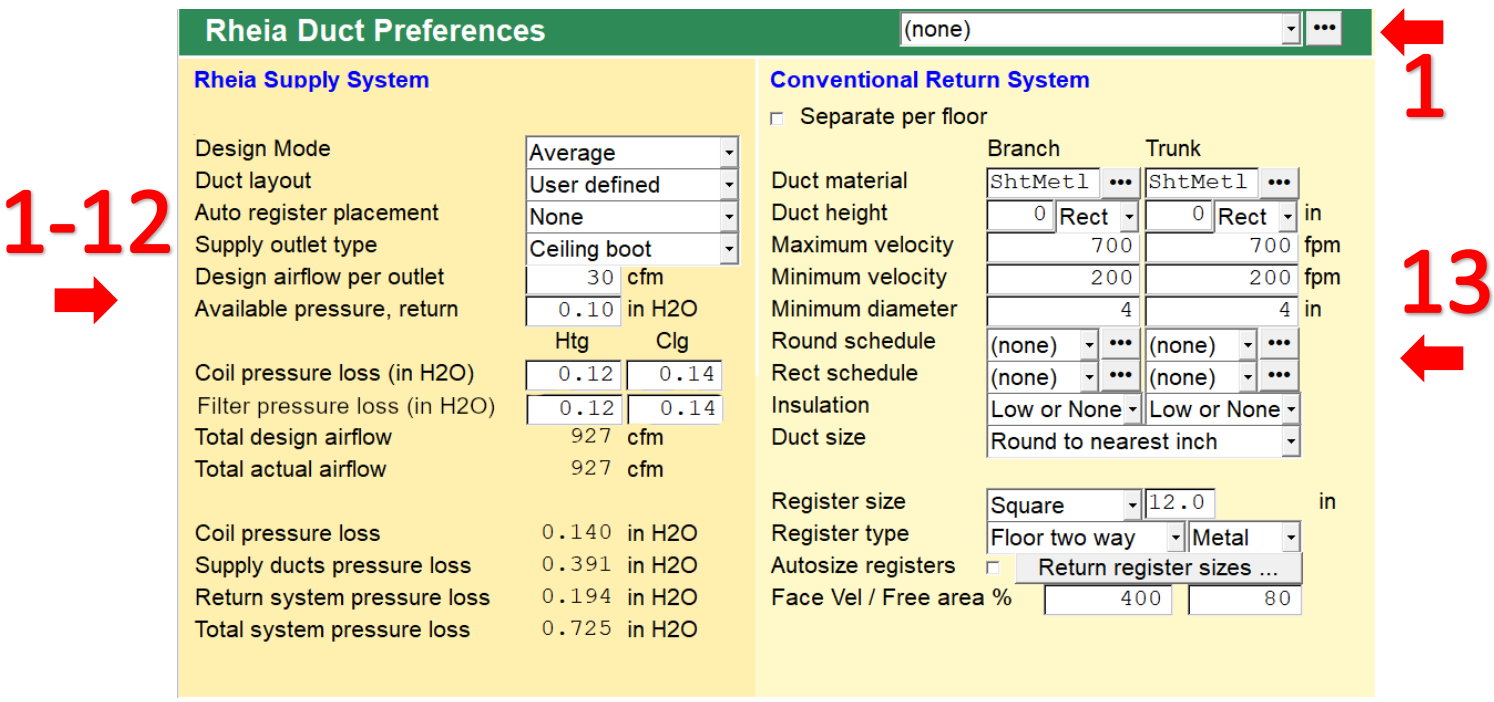


8. Go to the 'Show' menu and select the 'Right-D" option, then select the 'Preferences' option. Select a Rheia duct library at the top right corner of the page (see Section 2 for setup instructions).
9. Go to the 'File' menu and select the 'save as' option and choose the 'template' option (you can choose the default template option and skip the next step if the majority of your designs will be Rheia designs). Enter a name for this template like 'Rheia Designs' that tells can identify as the template to start with when doing a Rheia design.

Note: If you do not start your job with your 'Rheia Template' from above, you can make the above changes to a project after the fact.

Section 2- Preferences setup

1. Create a library. A library will remember your settings and can be selected in any job allowing you to reuse the information you enter on this page and reduce design steps for future use. To create a library, click the button labeled 1 in the figure below. Create a new library by clicking the 'new' button and name your library ('Rheia Design Preferences' for example). Make sure to select the 'Rheia' design option before clicking okay. Changes can also be made directly to the preferences page.



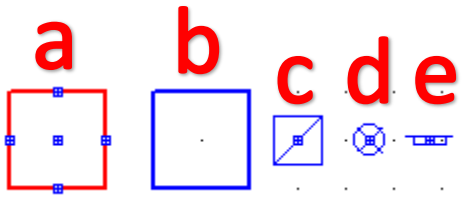
2. Design Mode: This setting determines how the CFM will be set. Cooling will use the cooling load to determine airflow. Heating will use the heating load to determine airflow. Average will use an average of the two.
3. Duct layout: Rheia duct systems have a unique layout. All systems will need to be user defined before they are completed. See section 3 for more information on laying out the duct system.
4. Auto register placement: Can be used to automatically place registers. Originally designed for traditional duct systems, adjustments will be necessary. NOTE: if used, this setting must be returned to 'none' in order to manually adjust register placement.
5. Supply outlet type: Choose the outlet type you will use most: Ceiling Boot, Sidewall Slot Boot, or Pass-through Slot boot. These can be changed on the drawing screen. See the Rheia design guide for more information on boot types.
6. Design airflow per outlet: Enter the baseline target airflow per outlet. This will determine the starting number of outlets a room will have and serve as a baseline for the initial balance of the system. Actual CFM will be calculated at each outlet as changes are made to the design.

7. Available pressure, return: Enter the target amount of pressure loss for which the return system will be designed.
8. Coil pressure loss: Enter the pressure loss rating for your coil. This will be the wet number in most climates for cooling and the dry for some very dry climates in cooling and all climates in heating. Many* fan coils include the coil pressure drop in their ratings but may have a wet coil adjustment that should be entered into the Clg field. *This excludes some modular fan coils and is determined by each manufacturer, for accuracy please consult manufacturer data.
9. Filter pressure loss: Enter the pressure loss across the filter with which the system is being designed.
10. Total design airflow: The airflow the system is being designed to achieve. This is based on equipment information and the design mode above.
11. Total actual airflow: The airflow the design was actually able to achieve through the user's design and automated balancing calculations.
12. Coil, Supply, Return, and Total pressure loss: The actual pressure loss values of the user's design. Used to check the efficacy of the design.
13. Conventional Return System Preferences: Duct preference settings for the conventional return. See Mitek/Wrightsoft's standard documentation.
14. Conventional Fitting Preferences: Duct fitting preference settings for the conventional return. See Mitek/Wrightsoft's standard documentation.

Section 3- Drawing the duct system

Prepare the drawing screen

1. Go to the drawing screen and check the building, duct, and duct notation layers for the floor you want to stat the design. You can start on any floor you choose, but it is easiest to start one floor at a time.
2. Identify the key duct components:
    a. The air handler (furnace/fancoil)
    b. The Rheia Manifold
    c. Return grill (s)
    d. Rheia ceiling boot
    e. Rheia sidewall slot boot or pass-through slot boot



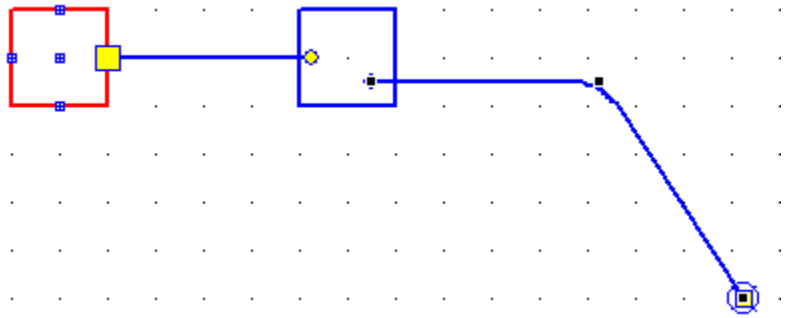
3. Drag and drop these components to the locations you plan to install them.
4. Add and locate supply and return outlets and grills as needed/desired.
5. Right click on the grills and outlets to adjust size and type.
6. Repeat this process for each floor of the building.
7. Turn on all floors of the building to try to identify conflicts in your layout.

Drawing the duct

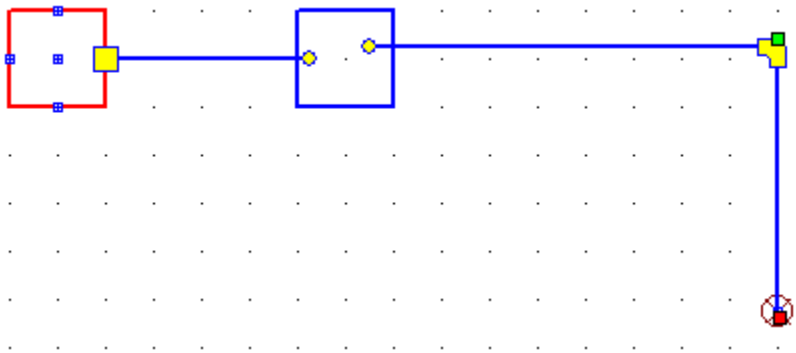
1. The return system connects normally. See Mitek/Wrightsoft's standard support documentation.
2. The supply side must start at an air handler snap and connect a duct to the Rheia manifold**. The Rheia manifold does not require the use of snap points**. Connect the duct to any point on the manifold as seen if figure to the right. See the Rheia design guide for best practices for connecting the air handler to the manifold.

3. Strech the manifuld as needed. See Rheia design guides for proper sizing of the manifold.
4. Once the manifold is connected to the air handler and the manifold is sized to your specifications, start connecting the manifold to the supply resisters.
    a. **All supply branches in Rheia duct systems are home-run**. There are no 'wye' or 'tee' style connections. Each supply outlet has iits own duct connecting it to the manifuld.

b. If the supply line will be run without any 45 or 90 connectors, draw a straight-line duct from the manifold to the outlet. To reroute the path of the duct, make sure it's selected and click on the edit-points icon: found at the top left tool bar on the drawing screen. Use the edit points to bend the path of the duct. See the Rheia design guide for guidelines on bending duct vs using 90 and 45 connectors.



c. If you intend to use a 90 or 45 connector in your supply run, draw a straight-line duct to the place you intend to put the connector. Then picking up from the end of that duct draw another straight-line just as you would with a standard duct design.



d. If your system requires risers to connect the manifold to the outlets follow the same procedures you would follow for standard duct. Add a fitting to the manifold from the HVAC shapes if the riser is coming directly off of the manifold, otherwise draw a duct to the point where the riser will be. On the property sheet for the fitting where your riser should be, change the 'Is a riser?' option to 'yes' then enter the vertical length of the riser. Because the Rheia ducts are flexible, you will also need to identify the elbow location(s) to indicate whether you will have an elbow at the top of the riser, at the bottom, both, or none (using bends instead).

e. Connect the return as you would a standard duct system in RSU.

Automated duct draw

If your duct system is not going to use risers or many elbows, you may find it fastest to right click the air handler and set the 'duct layout' option to 'radial' so that RSU will make all of the duct connections as straight line ducts. You can then immediately return the 'duct layout' setting to 'user defined' and use the edit points to adjust the paths of each run. NOTE: on more complicated systems this method tends to be more work than drawing from scratch.

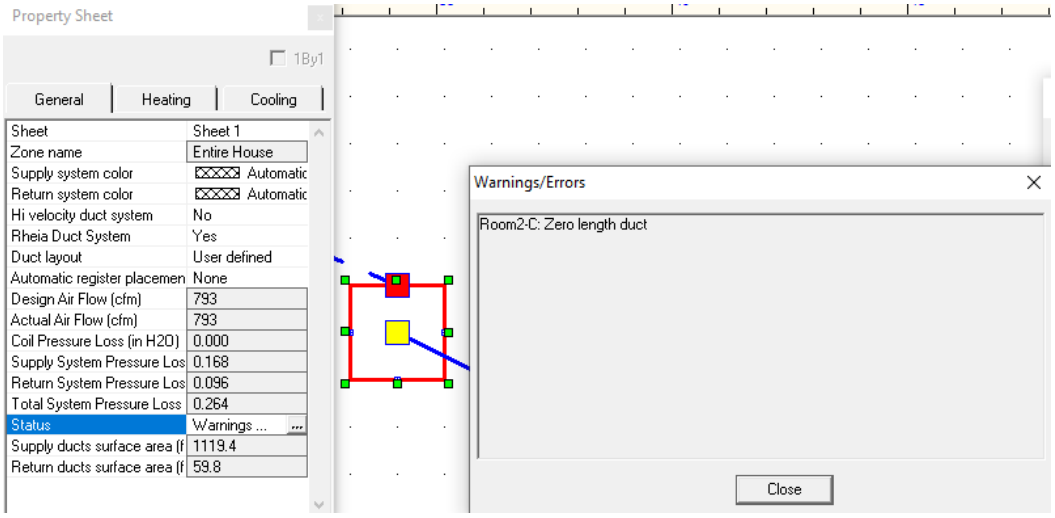
Section 4- Checking the duct system

Follow Rheia's design guidelines.

The drawing screen

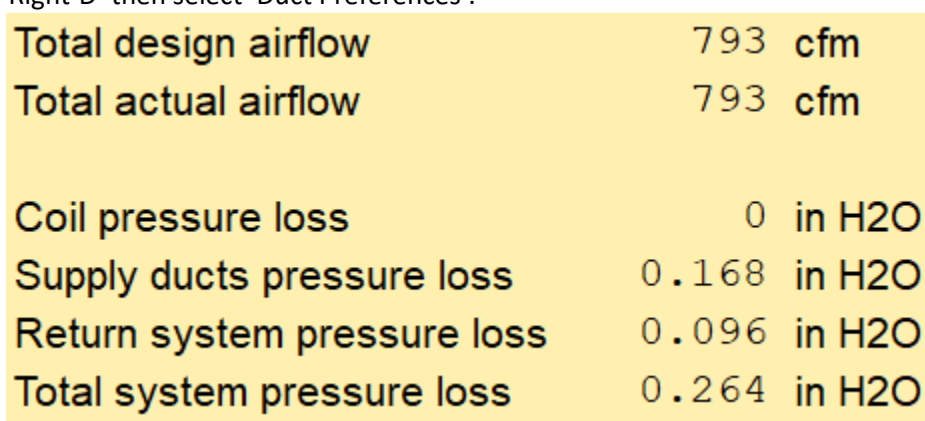
1. Review the duct drawing. Make sure you have enough space where ducts are bunched together. Try looking at the system one sheet at a time to make sure your ducts are following the path you want.

2. Look at the air handler. If it is a bright red color, that indicates that there are design warnings. Right click it and go to the general tab of the air handler property sheet. Look for the 'status' field. If the air handler is bright red, you will likely see the word 'warnings' in this field. Click in this field and read the warnings/errors window to identify what might be wrong with the design. Adjust the design to resolve the warnings. For example, if you see a 'zero duct length' message there is an outlet without a duct attached to it and attached to the manifold- find where it is disconnected and attach it.



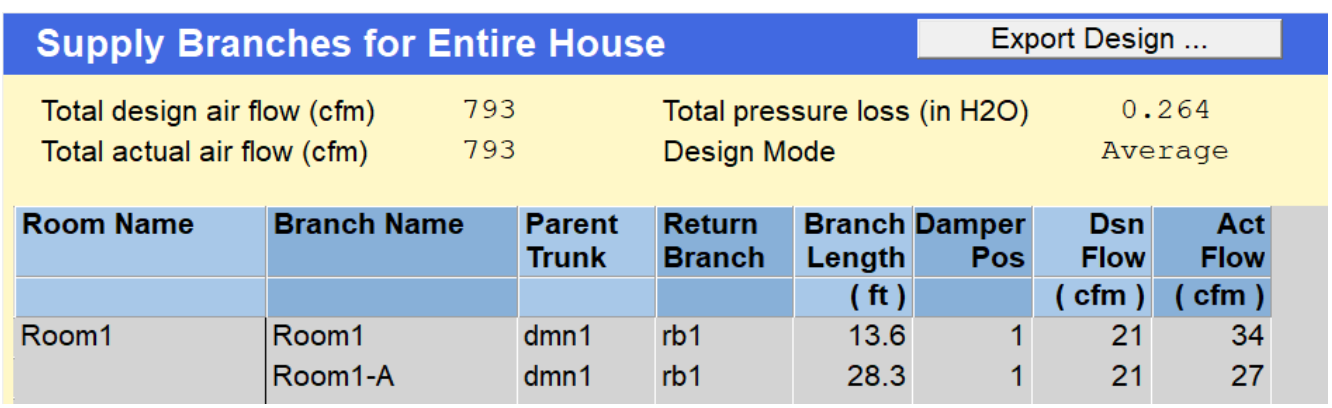
The duct preferences page.

1. Go to the 'show' menu and select 'Right-D' then select 'Duct Preferences'.
2. Here you can see the pressure loss on the system broken into supply and return. Follow Rheia guidelines for proper pressure ranges.



The supply branches page.

1. Go to the 'show' menu, select the 'Right-D' option, and select the 'supply branches' option.



Room Name	Branch Name	Parent Trunk	Return Branch	Branch Length	Damper Pos	Dsn Flow	Act Flow
				(ft)		(cfm)	(cfm)
Room1	Room1	dmn1	rb1	13.6	1	21	34
	Room1-A	dmn1	rb1	28.3	1	21	27

2. The most useful fields on this page are the far-right columns. Here you can see the branch length and damper position of each run and compare design (target) airflow with the actual airflow. You can use this page as well as the 'show longest ducts' option on the drawing screen to troubleshoot and adjust a design.
3. Once your design is complete you can export the design information for the Rheia Portal using the 'Export Design' button in the top right corner. Choose a location and name for the file. This file can then be manually imported to Rheia's online portal.